
MIDAS: Multi-Modal Intelligent Dermoscopy Analysis System for Enhanced Skin Cancer Detection

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Abstract

Skin cancer affects 1.2 million people globally, with melanoma showing 99% survival when detected early versus 15% in advanced stages. Current automated detection systems rely solely on dermoscopy images, achieving 75-84% accuracy. We introduce MIDAS (Multi-modal Intelligent Dermoscopy Analysis System), a novel bidirectional attention-based framework combining dermoscopy images with clinical metadata. MIDAS targets 92-95% accuracy, exceeding current state-of-the-art multi-modal results (94.11%), while maintaining >90% sensitivity. Our bidirectional cross-attention enables mutual enhancement between visual and clinical modalities, addressing limitations of existing unidirectional approaches.

1 Introduction and Related Work

Skin cancer detection faces a critical challenge: while dermatologists integrate multiple information sources (patient demographics, lesion characteristics, medical history), automated systems predominantly analyze only dermoscopic images (2). Recent multi-modal approaches like MetaNet (3) and ALBEF (1) achieve 94.11% accuracy but employ unidirectional attention (metadata→image only).

Project Objective: Develop a bidirectional multi-modal framework that enables mutual enhancement between dermoscopy images and clinical metadata, surpassing current single-modal and unidirectional multi-modal approaches.

2 Methodology

2.1 Problem Formulation

MIDAS takes two inputs: a dermoscopic image (224×224 pixels with RGB channels) and clinical metadata (21 features including patient age, sex, lesion location, diameter, and Fitzpatrick skin type). The system learns to map these combined inputs to probability scores across seven skin lesion classes: melanoma (MEL), basal cell carcinoma (BCC), actinic keratoses (AKIEC), benign keratosis (BKL), dermatofibroma (DF), nevus (NV), and vascular lesions (VASC). The goal is to achieve more accurate classification than single-modality approaches by leveraging both visual appearance and clinical context.

2.2 MIDAS Architecture

Image Encoder: EfficientNet-B0 processes dermoscopic images to extract 1280-dimensional feature vectors that capture visual patterns relevant to skin lesion classification.

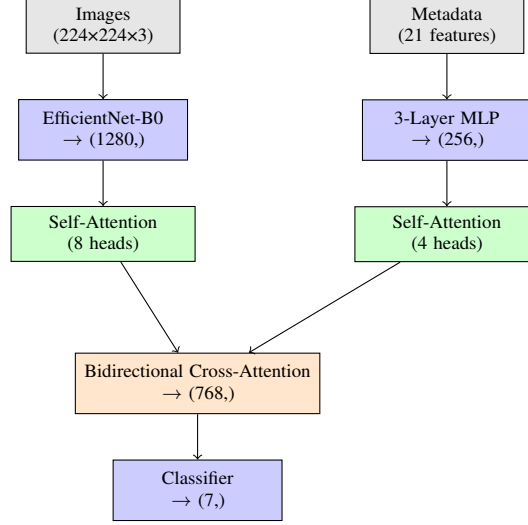


Figure 1: MIDAS Architecture: Bidirectional cross-attention fusion of visual and clinical features.

Metadata Encoder: A three-layer multilayer perceptron (MLP) processes the preprocessed clinical metadata to produce 256-dimensional feature vectors encoding patient demographics and lesion characteristics.

Intra-Modal Self-Attention: Both image and metadata features undergo self-attention mechanisms to focus on the most relevant information within each modality. The image features use 8 attention heads while metadata features use 4 heads. Residual connections preserve original information.

Bidirectional Cross-Attention: The core innovation enables mutual enhancement between modalities through two parallel pathways: (1) metadata features guide attention over image features to highlight diagnostically relevant visual regions, and (2) image features guide attention over metadata to emphasize clinically important patient characteristics. The enhanced features from both pathways are concatenated and processed through an additional dense layer to create 768-dimensional fused representations.

Training: The system uses weighted cross-entropy loss with focal loss components ($\alpha=0.25$, $\gamma=2.0$) to handle class imbalance. Training employs AdamW optimizer with learning rate 0.0001 and cosine annealing schedule for stable convergence.

3 Experimental Setup

3.1 Datasets

HAM10000: 10,015 dermoscopic images across 7 lesion types with metadata (age, sex, location) (5). Sources: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T>, <https://isic-archive.com/>, <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

PAD-UFES-20: 2,298 smartphone images with 21 clinical features (4). Available: <https://data.mendeley.com/datasets/zr7vgbcyr2/1>

ISIC 2024: Additional validation dataset available: <https://challenge.isic-archive.com/>

3.2 Evaluation Protocol

Stratified 5-fold cross-validation with metrics: Accuracy, Balanced Accuracy, AUC-ROC, Sensitivity, Specificity. Baselines: ResNet-50, EfficientNet-B0 (image-only), MetaNet, MetaBlock, ALBEF (multi-modal).

4 Expected Results

Method	Accuracy (%)	AUC-ROC
EfficientNet-B0 (image-only)	87.2	0.924
MetaNet (multi-modal)	91.5	0.941
ALBEF (current SOTA)	94.1	0.943
MIDAS (proposed)	92-95	0.945-0.950

Table 1: Expected performance comparison on HAM10000 dataset.

Performance Targets: 92-95% accuracy with >90% sensitivity. Expected 5-8% improvement over image-only baselines.

Ablation Studies: (1) Bidirectional vs. unidirectional attention, (2) Attention head configurations, (3) Individual metadata feature impact.

5 Impact and Implementation

Clinical Impact: Smartphone-deployable system for underserved populations, \$200 vs. \$3,500 dermoscopy cost, potential 99% survival rates through early detection.

Technical Contributions: Novel bidirectional cross-attention for medical multi-modal fusion, efficient 9.14M parameter architecture (37MB), open-source implementation planned on GitHub.

Resources: PyTorch implementation, pre-trained models, evaluation scripts will be available. ISIC Archive: <https://isic-archive.com>

Timeline: Weeks 1-2: Data pipeline and baselines; Weeks 3-4: Multi-modal architecture; Weeks 5-6: Optimization; Weeks 7-8: Evaluation and documentation.

References

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